Rethinking ABM Architecture

Entity Component Systems and the Case for Parallel Agent Interaction

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Outline

1. Cars driving in circles 3
   1.1 The original model 4
   1.2 Overview 5
   1.3 The real world 6
2. ABM layouts 7
   2.1 Traditional ABM layout . . 8
   2.2 What if? 9
3. ECS (Entity Component System)! 10
   3.1 The history of ECS 11
   3.2 A short introduction 12
   3.3 Data layout in a classic ABM 13
   3.4 ECS data layout 14
   3.5 ABM in ECS terms 15
4. Cars 2 16
   4.1 Reimplementing the model using ECS 17
   4.2 Prediction strategies 18
   4.3 Which strategy works best 19
   4.4 The formation of habit . . 20
5. Appendix 21

— Outline

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5.1 Appendix 22
Bibliography 22

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1. Cars driving in circles

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1.1 The original model

- Influential paper by Hodgson (2006)

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1. Cars driving in circles
— 1.1 The original model

1.2 Overview

- Cars drive sequentially either clockwise or counterclockwise on a 100×2 ring and decide in each step, whether to turn take the left or right lane based on the decision formula

$$\mathrm{LR}^n = w_s S_s^n (2s^n - 1) + w_o S_o^n (2o^n - 1) + w_a S_a^n (c_r^n - c_l^n) + w_h S_h^n h^n.$$

  where s, o and c_r/c_l are determined by the cars ahead of car n and h^n is the habit of the driver.
- Driving on one lane shifts their habit towards that lane based on the update formula

$$h^n = h^n_{\text{prev}} + \frac{\text{lane}}{K + t}$$

1. Cars driving in circles
— 1.2 Overview

1.3 The real world

- Real-world agents (especially cars) operate in parallel.
- Why did Hodgson (2006) choose a sequential model?

1. Cars driving in circles
— 1.3 The real world

1. Ask why abs enforce sequential logic that is and that we have to go into more detail

1.3 The real world

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- Why did Hodgson (2006) choose a sequential model?
  - Because it is simple?
  - --→ In the classic ABM framework the sequential approach is the most natural approach.

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1.3 The real world

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- Why did Hodgson (2006) choose a sequential model?
  - Because it is simple?
  - --→ In the classic ABM framework the sequential approach is the most natural approach.
  - But why does the sequential approach lend itself so well for these models?

1. Cars driving in circles
— 1.3 The real world

1. Ask why abs enforce sequential logic that is and that we have to go into more detail

2. ABM layouts

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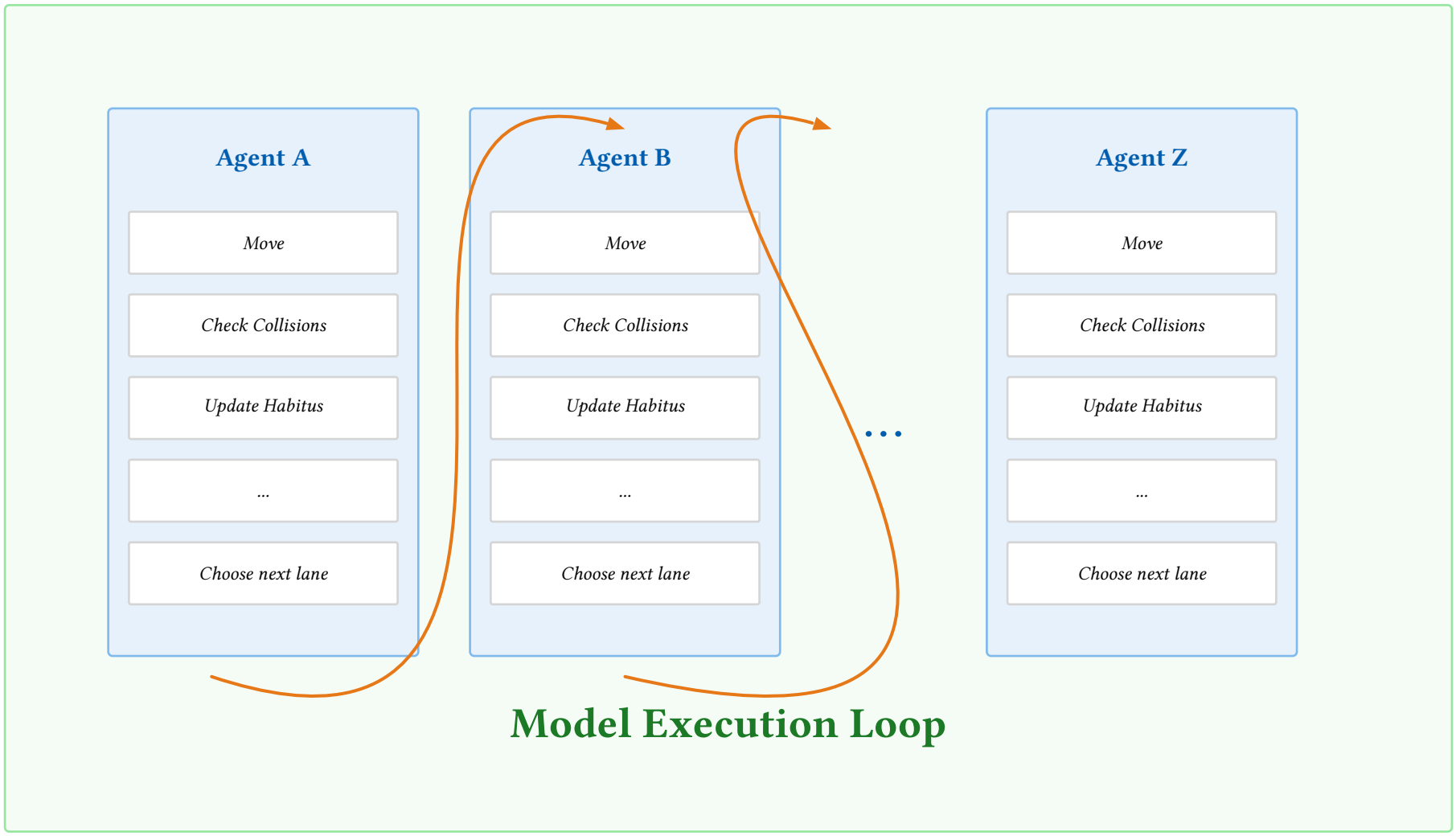
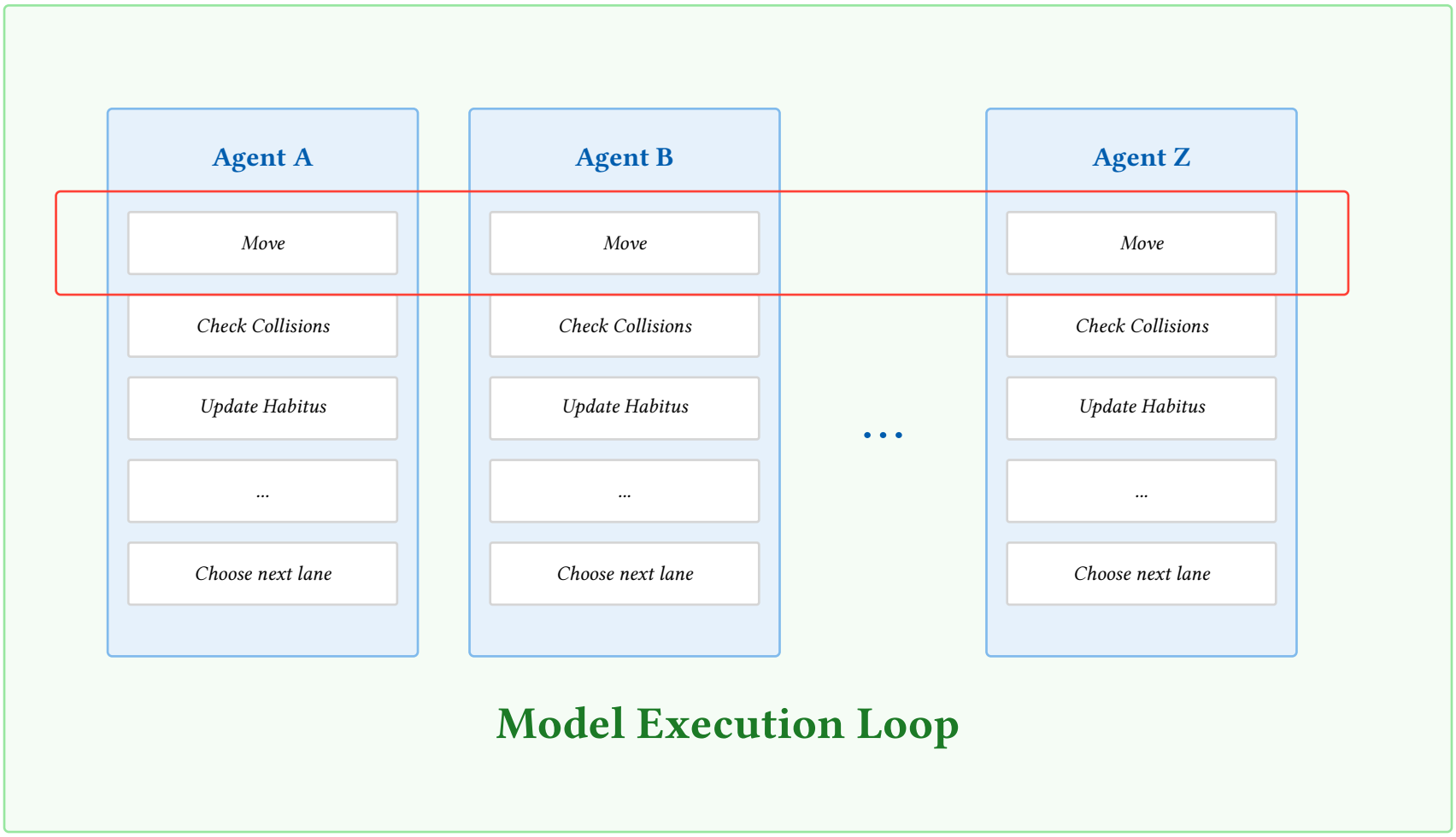
2.1 Traditional ABM layout



2. ABM layouts
— 2.1 Traditional ABM layout

1. Agents have step functions
2. these step functions run sequentialy

2.2 What if?



2. ABM layouts
— 2.2 What if?

1. What if we group the data not by agents but by systems
2. Luckily paradigm exists => ECS

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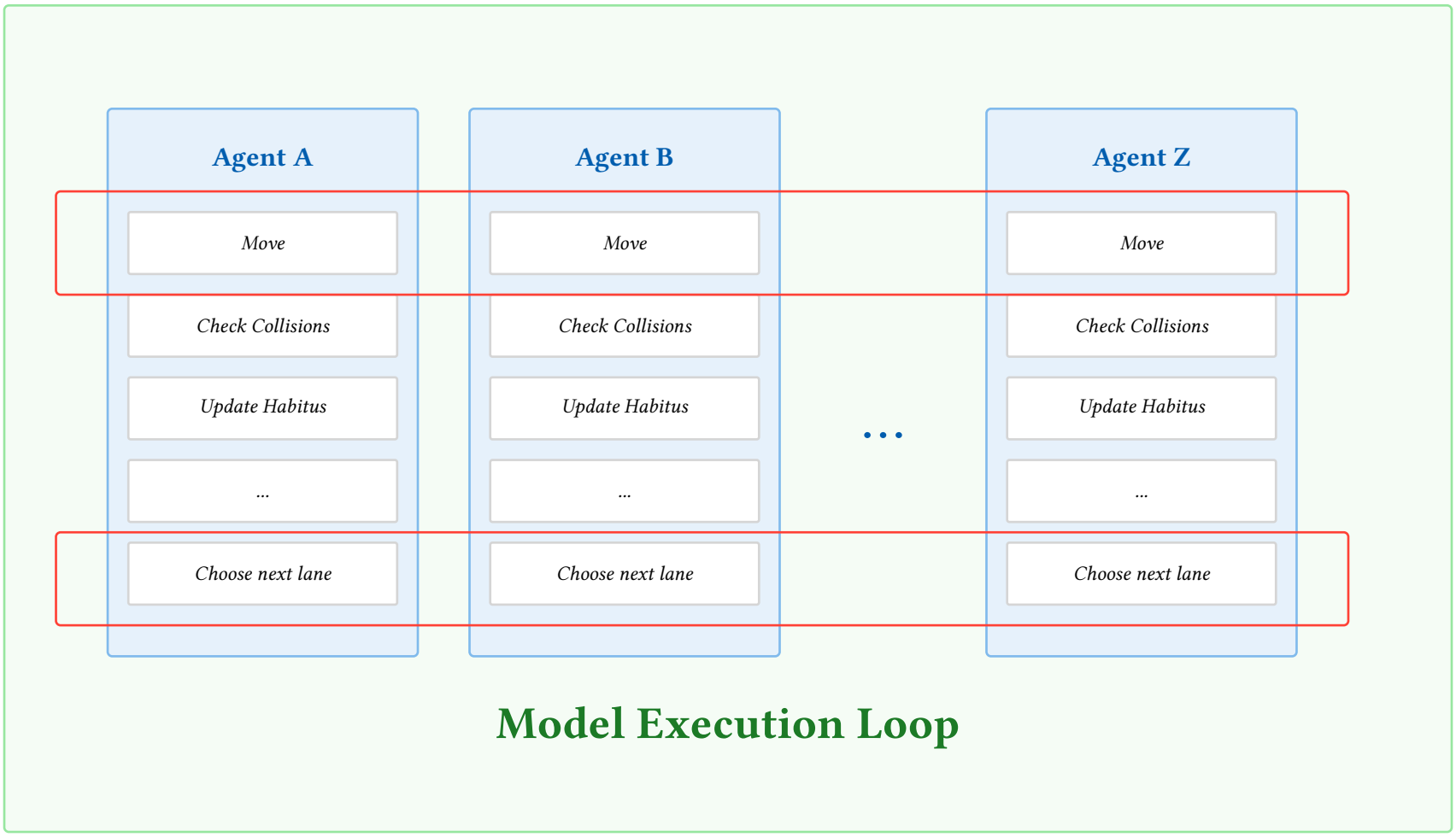

1. What if we group the data not by agents but by systems
2. Luckily paradigm exists => ECS

3. ECS (Entity Component System)!

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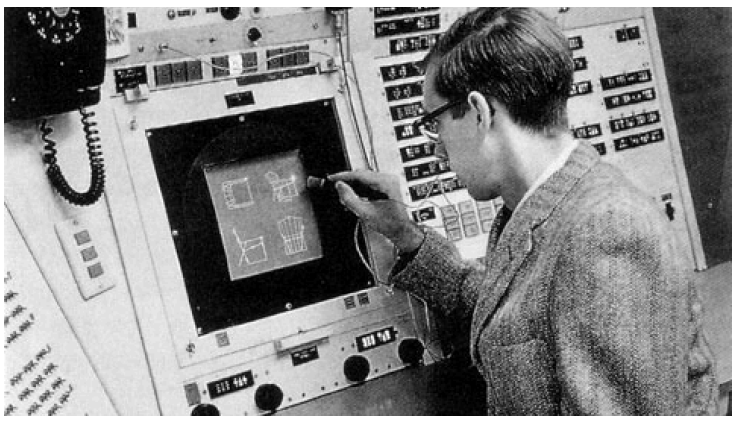
3.1 The history of ECS

- The origins of ECS reach back to 1959 where Ivan Sutherland pioneered it in a drawing program,one of the first graphical user interfaces (Sutherland, 2003).
- Now ECS is primarily used in game development. (e.g Bevy Engine, Unity Dots, ...)



3. ECS (Entity Component System)!
— 3.1 The history of ECS

3.2 A short introduction

- ECS is a way to separate data from functionality
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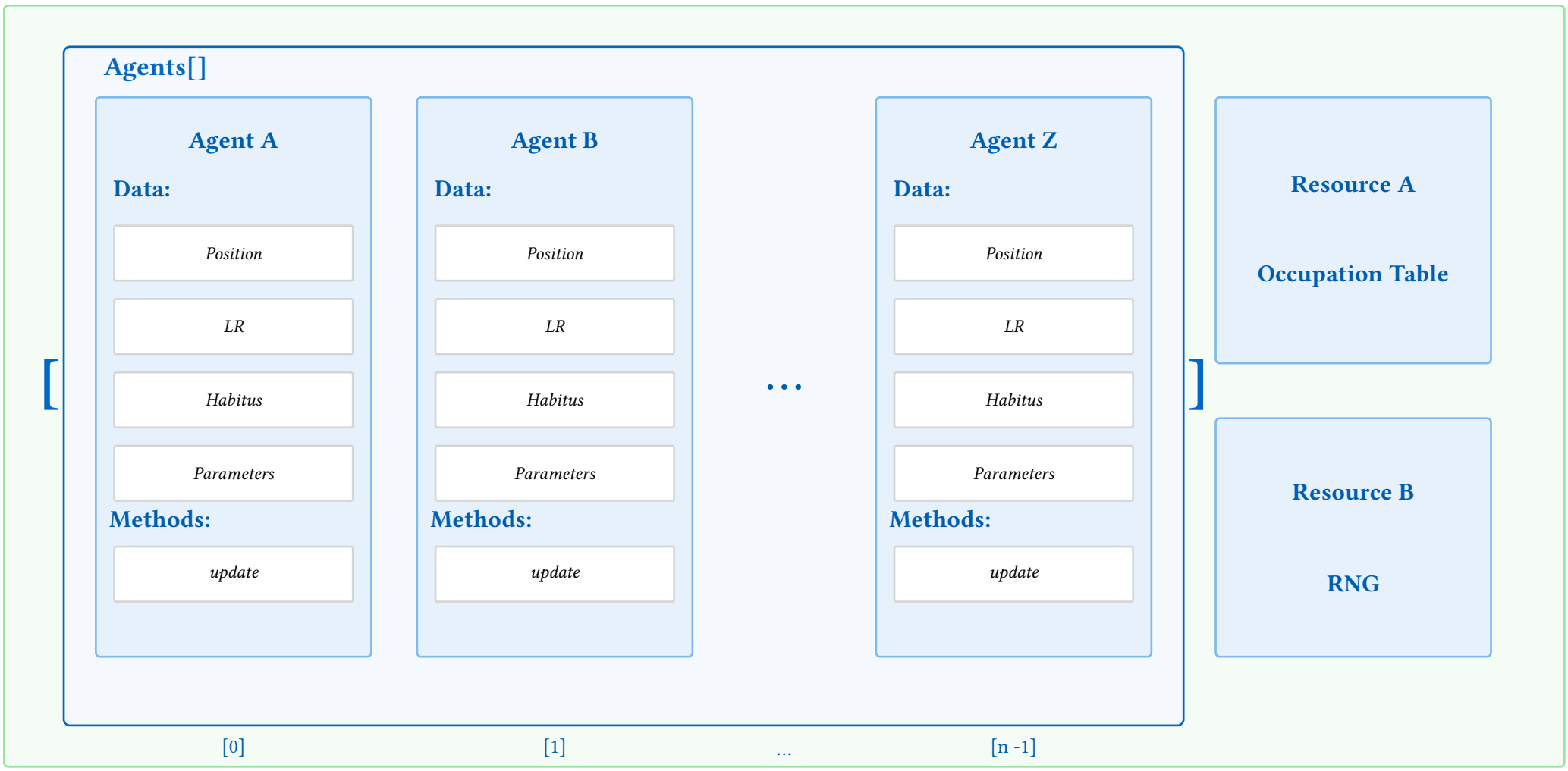
3. ECS (Entity Component System)!
— 3.2 A short introduction

3.3 Data layout in a classic ABM 3. ECS (Entity Component System)!



3. ECS (Entity Component System)!
— 3.3 Data layout in a classic ABM

3.4 ECS data layout

	Entity 1	Entity 2	Entity 3
Component 1	*[Data]*		
Component 2	*[Data]*	*[Data]*	*[Data]*
Component 3		*[Data]*	*[Data]*

3. ECS (Entity Component System)!
— 3.4 ECS data layout

3.5 ABM in ECS terms

Concept	Description	ABM Equivalent	Role in Simulation
Entity	Unique object in the world	Agent	Represents an individual actor in the simulation
Component	Data attached to an entity	Agent attributes / state variables	Stores properties such as position, preferences, or resources
System	Function operating on sets of components	Part of the agent step function	Implements simulation rules and updates state

4. Cars 2

4. Cars 2

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4.1 Reimplementing the model using ECS

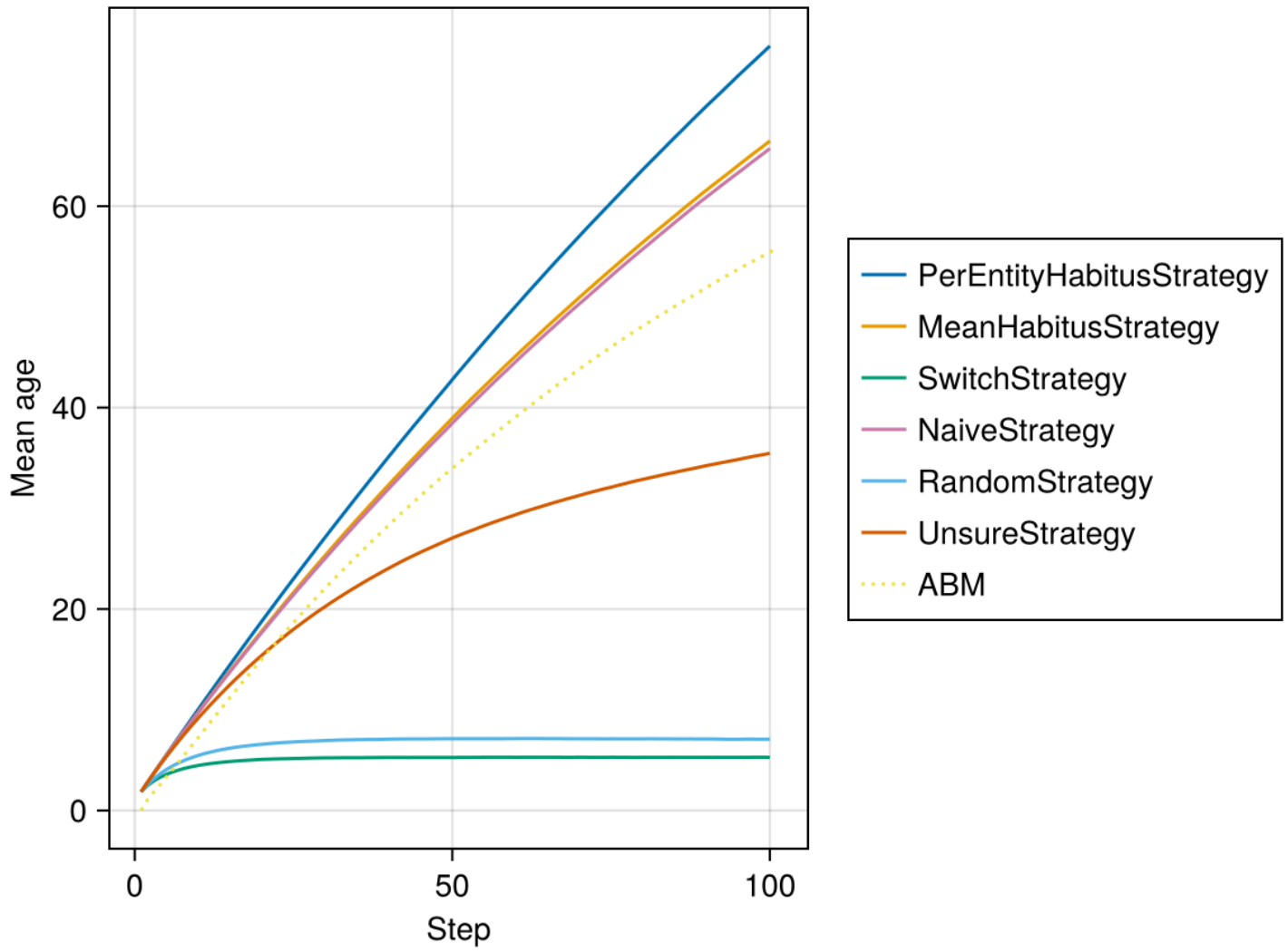
- The natural way to implement the model in ECS is so that cars move simultaneously.
- This leads to a problem:
  - How does a car predict where the other cars are in the next step?
  - --→ different prediction strategies.

4. Cars 2
— 4.1 Reimplementing the model using ECS

4.2 Prediction strategies

Strategy	Prediction
Naive strategy	Other cars just stay on their lane
Switch strategy	Other cars switch their lane
Unsure strategy	Both lanes are equally likely
Per Entity Habitus	The car switches lanes based on its habitus
Mean Absolute Habitus	The car is more likely to stay on a lane the higher the mean absolute habitus is
Random strategy	The car gets assigned a random probability to switch lane

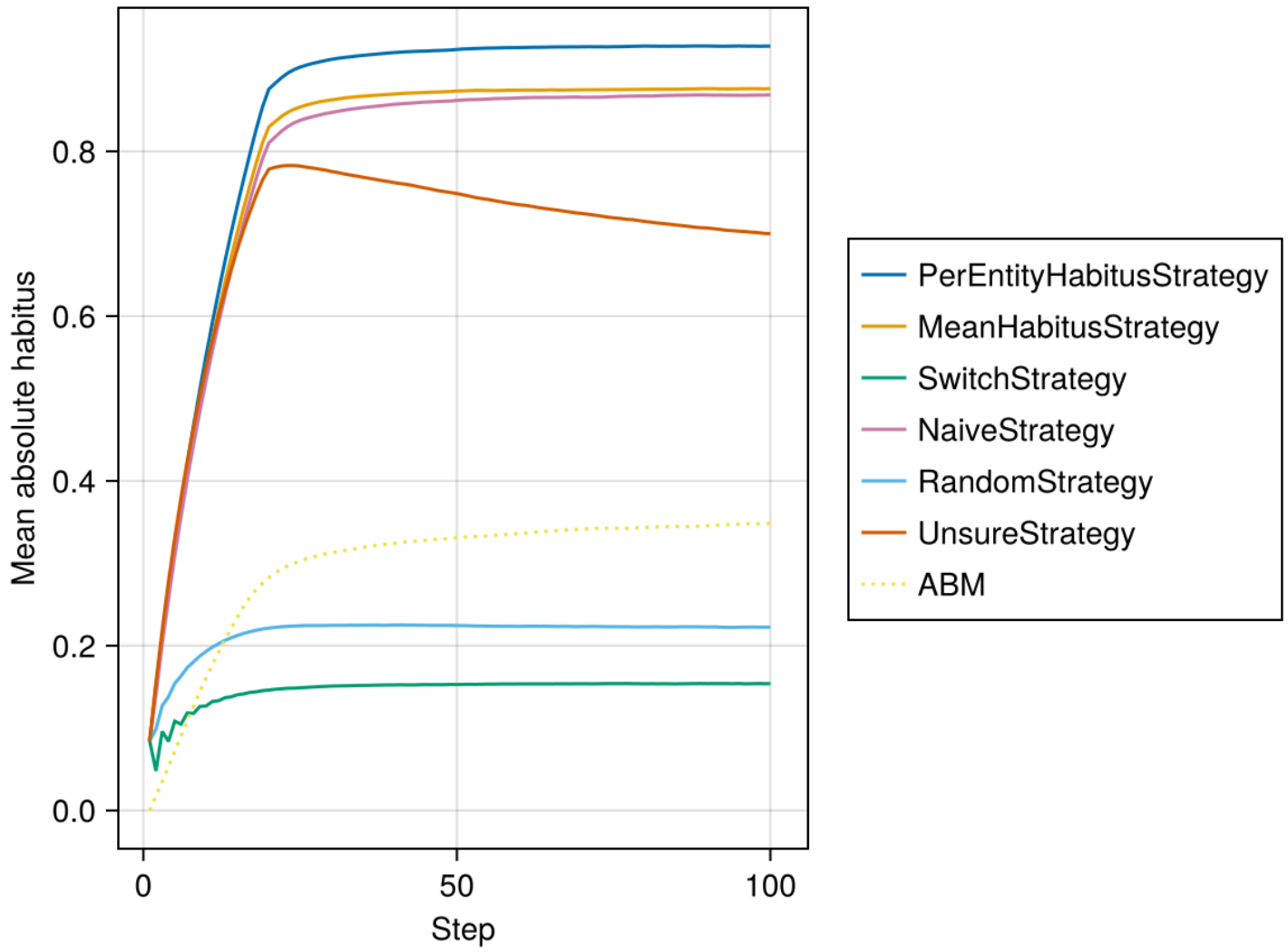
4.3 Which strategy works best



4. Cars 2

— 4.3 Which strategy works best

4.4 The formation of habit



5. Appendix

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